

Exercise of Supervised Learning: Multiclass Classification

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Exercise 4: Multiclass Hinge Loss

Consider the multiclass classification scenario consisting of a feature space $\mathcal{X}$ and a label space $\mathcal{Y} = \{1, \dots, g\}$ with $g \geq 2$ classes. Moreover, we consider the hypothesis space of models based on g discriminant/scoring functions:

$$\mathcal{H} = \{f = (f_1, \dots, f_g)^T : \mathcal{X} \rightarrow \mathbb{R}^g \mid f_k : \mathcal{X} \rightarrow \mathbb{R}, \forall k \in \mathcal{Y}\}.$$

A model f in $\mathcal{H}$ is used to make a prediction by means of transforming the scores into classes by choosing the class with the maximum score:

$$h(\mathbf{x}) = \arg \max_{k \in \{1, \dots, g\}} f_k(\mathbf{x}). \quad (1)$$

The multiclass hinge loss is defined by

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + \mathbf{1}_{y \neq k}).$$

(a) Show that 0-1-loss for a predictor h as in (1) based on a model $f \in \mathcal{H}$ is at most the multiclass hinge loss for f i.e.,

$$L_{0-1}(y, h(\mathbf{x})) = \mathbf{1}_{y \neq h(\mathbf{x})} \leq L(y, f(\mathbf{x})).$$

Digest the loss & roadmap for (a)

Read the hinge loss term by term:

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k}).$$

- ▶ $f_k(\mathbf{x}) - f_y(\mathbf{x})$: score gap of k over true class y .
- ▶ $I_{y \neq k}$: +1 if $k \neq y$, 0 if $k = y$.
- ▶ $\max_k$: take the worst offender.
- ▶ $k = y$ gives term = 0, so $L \geq 0$ *always*.

What (a) asks. 0-1 loss is 0 (correct) or 1 (wrong). Want $L(y, f(\mathbf{x})) \geq L_{0-1}(y, h(\mathbf{x}))$ in both regimes.

Roadmap: split by whether $\arg \max$ **matches** y .

- ▶ Case 1, $y = \arg \max_k f_k(\mathbf{x})$: pick the trivial term $k = y$ inside the max; compare against L_{0-1} .
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Solution to (a) — Case 1: $y = \arg \max_k f_k(\mathbf{x})$

Prediction correct, so

$$L_{0-1}(y, h(\mathbf{x})) = 0.$$

Plug $k = y$ into the max:

$$f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k} = 0 - 0 + 0 = 0.$$

Hence

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k}) \geq 0.$$

$$\Rightarrow L(y, f(\mathbf{x})) \geq 0 = L_{0-1}(y, h(\mathbf{x})). \quad \checkmark$$

0-1 loss

$$L_{0-1}(y, h(\mathbf{x})) = I_{y \neq h(\mathbf{x})}$$

Multiclass hinge

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Solution to (a) — Case 2: $y \neq \arg \max_k f_k(\mathbf{x})$

Prediction wrong, so

$$L_{0-1}(y, h(\mathbf{x})) = 1.$$

Pick $k^* = \arg \max_k f_k(\mathbf{x})$. Then $f_{k^*}(\mathbf{x}) > f_y(\mathbf{x})$ and $I_{y \neq k^*} = 1$:

$$\underbrace{f_{k^*}(\mathbf{x}) - f_y(\mathbf{x})}_{>0} + \underbrace{I_{y \neq k^*}}_{=1} > 1.$$

Hence

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k}) > 1.$$

$$\Rightarrow L(y, f(\mathbf{x})) > 1 = L_{0-1}(y, h(\mathbf{x})). \quad \checkmark$$

Both cases combined: $L_{0-1}(y, h(\mathbf{x})) = I_{y \neq h(\mathbf{x})} \leq L(y, f(\mathbf{x})).$

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Question (b)

(b) Verify that the multiclass hinge loss of $f \in \mathcal{H}$ on a data point $(\mathbf{x}, y) \in \mathcal{X} \times \mathcal{Y}$ is bounded from above by $\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$. *Hint:* Note that this upper bound is sometimes referred to as the multiclass hinge loss.

Digest (b) & how to reuse (a)

Difference from (a).

- ▶ (a): hinge upper-bounds 0-1.
- ▶ (b): *sum surrogate* upper-bounds hinge.
- ▶ Same direction; swap $\max_k$ for $\sum_{k \neq y} \max\{0, \cdot\}$.

Reuse (a)'s case split.

- ▶ Same case split as (a), via $k^* = \arg \max_k f_k(\mathbf{x})$.
- ▶ From (a): in case 2, the max in L is attained at k^* .
- ▶ That k^* -summand already sits inside RHS; the remaining summands are ≥ 0 , so $\text{RHS} \geq L$.

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Sandwich (a)+(b)

$$L_{0-1} \stackrel{(a)}{\leq} L \stackrel{(b)}{\leq} \sum_{k \neq y} \max\{0, \cdot\}$$

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If $k^* = y$, plug $k = k^* = y$ into the max:

$$L(y, f(\mathbf{x})) = f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 0 = 0.$$

RHS is a sum of $\max\{0, \cdot\} \geq 0$, hence non-negative:

$$L(y, f(\mathbf{x})) = 0 \leq \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}. \quad \checkmark$$

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Solution to (b) — Case 1: $y = k^*$

If $k^* = y$, plug $k = k^* = y$ into the max:

$$L(y, f(\mathbf{x})) = f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 0 = 0.$$

RHS is a sum of $\max\{0, \cdot\} \geq 0$, hence non-negative:

$$L(y, f(\mathbf{x})) = 0 \leq \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}. \quad \checkmark$$

Multiclass hinge

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k})$$

Sum surrogate

$$\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$$

Solution to (b) — Case 2: $y \neq k^*$

From (a): $L(y, f(\mathbf{x})) = f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 1 > 1$.

$$\begin{aligned} L(y, f(\mathbf{x})) &= \underbrace{f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 1}_{\geq 0} \\ &= \max\{0, f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 1\} \\ &\leq \max\{0, f_{k^*}(\mathbf{x}) - f_y(\mathbf{x}) + 1\} \\ &\quad + \underbrace{\sum_{k \neq y, k^*} \max\{0, f_k(\mathbf{x}) - f_y(\mathbf{x}) + 1\}}_{\geq 0} \\ &= \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}. \quad \checkmark \end{aligned}$$

Both cases combined:

$$L(y, f(\mathbf{x})) \leq \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}. \quad \blacksquare$$

Multiclass hinge

$$L(y, f(\mathbf{x})) = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k})$$

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$$\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$$

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Question (c)

In the case of binary classification, i.e., $g = 2$ and $\mathcal{Y} = \{-1, +1\}$, we use a single discriminant model $f(\mathbf{x}) = f_1(\mathbf{x}) - f_{-1}(\mathbf{x})$ based on two scoring functions: $f_1, f_{-1} : \mathcal{X} \rightarrow \mathbb{R}$ for the prediction by means of $h(\mathbf{x}) = \text{sgn}(f(\mathbf{x}))$. Here, f_1 is the score for the positive class and f_{-1} is the score for the negative class. Show that the upper bound in (b) coincide with the binary hinge loss $L(y, f(\mathbf{x})) = \max\{0, 1 - yf(\mathbf{x})\}$.

Digest (c) & how to reuse (b)

Difference from (b).

- ▶ Specialize: $g = 2$, $\mathcal{Y} = \{-1, +1\}$.
- ▶ Single discriminant: $f(\mathbf{x}) = f_1(\mathbf{x}) - f_{-1}(\mathbf{x})$.
- ▶ Sum collapses to one $\max\{0, \cdot\}$ term (only one wrong class).
- ▶ Target: that term = $\max\{0, 1 - yf(\mathbf{x})\}$, the binary hinge.

Reuse (b)'s surrogate.

- ▶ Start from $\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$; only one term survives.
- ▶ Split by $y = +1$ vs. -1 to identify the wrong class.
- ▶ Substitute $f_1 - f_{-1} = f(\mathbf{x})$ to fold into $1 - yf(\mathbf{x})$.

Starting point (from (b))

$$\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$$

Target: binary hinge

$$\max\{0, 1 - yf(\mathbf{x})\}$$

Binary setup

$$\begin{aligned} g &= 2, \mathcal{Y} = \{-1, +1\} \\ f(\mathbf{x}) &= f_1(\mathbf{x}) - f_{-1}(\mathbf{x}) \\ h(\mathbf{x}) &= \text{sgn}(f(\mathbf{x})) \end{aligned}$$

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Solution to (c) — Case 1: $y = +1$

$$\begin{aligned} & \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\} \\ &= \sum_{k \neq +1} \max\{0, f_k(\mathbf{x}) - f_1(\mathbf{x}) + 1\} \\ &= \max\{0, \underbrace{f_{-1}(\mathbf{x}) - f_1(\mathbf{x})}_{= -f(\mathbf{x})} + 1\} \\ &= \max\{0, 1 - f(\mathbf{x})\} \\ &= \max\{0, 1 - yf(\mathbf{x})\}. \quad \checkmark \end{aligned}$$

Starting point (from (b))

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Solution to (c) — Case 2: $y = -1$

$$\begin{aligned} & \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\} \\ &= \sum_{k \neq -1} \max\{0, f_k(\mathbf{x}) - f_{-1}(\mathbf{x}) + 1\} \\ &= \max\{0, \underbrace{f_1(\mathbf{x}) - f_{-1}(\mathbf{x})}_{= f(\mathbf{x})} + 1\} \\ &= \max\{0, 1 + f(\mathbf{x})\} \\ &= \max\{0, 1 - yf(\mathbf{x})\}. \quad \checkmark \end{aligned}$$

Both cases combined: sum surrogate = binary hinge
 $\max\{0, 1 - yf(\mathbf{x})\}$. ■

Starting point (from (b))

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Question (d)

Recall the statement of the lecture regarding the binary hinge loss:

“... the hinge loss only equals zero for a margin ≥ 1 encouraging confident (correct) predictions.”

Can we say something similar for the alternative multiclass hinge loss in (b)?

Hint: multiclass hinge loss: $\sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$, and all $\max\{0, \cdot\}$ terms are non-negative.

Think about:

- ▶ “Similar” = an iff for when the loss vanishes. Hypothesize: multiclass hinge = 0 $\iff$???
- ▶ Binary: 1 margin, condition “ ≥ 1 ”. Multiclass: how many margins per data point, and what replaces “ ≥ 1 ”?

Solution to Question (d)

Yes, we can say that *it is only zero if all the $g - 1$ margins are greater than 1*.

- ▶ Margins: $m_{y,k}(\mathbf{x}) = f_y(\mathbf{x}) - f_k(\mathbf{x})$, where $k \in \mathcal{Y} \setminus \{y\}$.
- ▶ Mathematically: $m_{y,k}(\mathbf{x}) \geq 1 \ \forall k \neq y \Leftrightarrow \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\} = 0$.

Proof:

$$\begin{aligned} m_{y,k}(\mathbf{x}) \geq 1 \ \forall k \neq y &\Rightarrow f_y(\mathbf{x}) - f_k(\mathbf{x}) \geq 1 \ \forall k \neq y \\ &\Rightarrow f_k(\mathbf{x}) - f_y(\mathbf{x}) \leq -1 \ \forall k \neq y \\ &\Rightarrow \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\} = 0 \ \forall k \neq y \\ &\Rightarrow \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\} = 0 \end{aligned}$$

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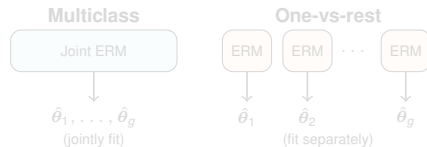
Question & Solution to (e): joint vs. one-vs-rest

Question. Linear scores $f_k(\mathbf{x}) = \theta_k^\top \mathbf{x}$. Compare two ERM approaches:

- ▶ ERM with the multiclass sum surrogate from (b).
- ▶ One-vs-rest: g independent binary-hinge ERMs.

Solution.

- ▶ Multiclass: *one* joint ERM minimising all $g - 1$ margins per data point *simultaneously*.
- ▶ One-vs-rest: g separate binary ERMs, each handling one margin (class vs. rest) in isolation.
- ▶ $\Rightarrow$ generally different parameters, different predictions.



Multiclass ERM (1 problem)

$$\min_{\theta_1, \dots, \theta_g} \sum_i \sum_{k \neq y_i} \max\{0, 1 + (\theta_k - \theta_{y_i})^\top \mathbf{x}^{(i)}\}$$

One-vs-rest ERM (g problems)

For each $c \in \mathcal{Y}$: $\min_{\theta_c} \sum_i \max\{0, 1 - y_i^{(c)} \theta_c^\top \mathbf{x}^{(i)}\}$, $y_i^{(c)} = +1$ if $y_i = c$, else -1 .

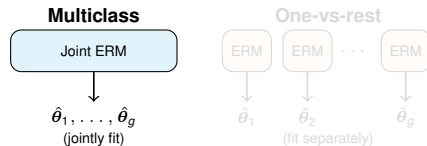
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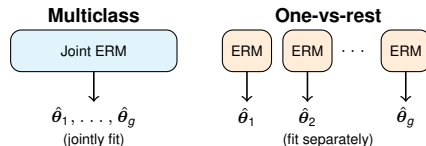
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For each $c \in \mathcal{Y}$: $\min_{\theta_c} \sum_i \max\{0, 1 - y_i^{(c)} \theta_c^\top \mathbf{x}^{(i)}\}$, $y_i^{(c)} = +1$ if $y_i = c$, else -1 .

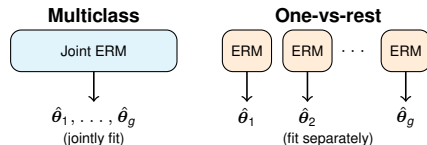
Question & Solution to (e): joint vs. one-vs-rest

Question. Linear scores $f_k(\mathbf{x}) = \theta_k^\top \mathbf{x}$. Compare two ERM approaches:

- ▶ ERM with the multiclass sum surrogate from (b).
- ▶ One-vs-rest: g independent binary-hinge ERMs.

Solution.

- ▶ Multiclass: *one* joint ERM minimising all $g - 1$ margins per data point *simultaneously*.
- ▶ One-vs-rest: g separate binary ERMs, each handling one margin (class vs. rest) in isolation.
- ▶ $\Rightarrow$ generally different parameters, different predictions.



Multiclass ERM (1 problem)

$$\min_{\theta_1, \dots, \theta_g} \sum_i \sum_{k \neq y_i} \max\{0, 1 + (\theta_k - \theta_{y_i})^\top \mathbf{x}^{(i)}\}$$

One-vs-rest ERM (g problems)

For each $c \in \mathcal{Y}$: $\min_{\theta_c} \sum_i \max\{0, 1 - y_i^{(c)} \theta_c^\top \mathbf{x}^{(i)}\}$, $y_i^{(c)} = +1$ if $y_i = c$, else -1 .

Code (Python): one-vs-rest

```
import numpy as np
from numpy.typing import NDArray
from scipy.optimize import minimize

# Vectorised empirical risk for binary hinge.
# Z = [X | y] of shape (n, p + 1); margins is (n,).
# All rows are handled in one matmul + elementwise op

def emp_risk_bin_hinge(theta: NDArray,
                      Z: NDArray) -> float:
    X, y = Z[:, :p], Z[:, p]
    margins = y * (X @ theta)
    return float(np.maximum(0.0, 1.0 - margins).sum())
```

```
# One-vs-rest fitting: g INDEPENDENT optims.
# Output: theta_mat of shape (p, g);
# column i = parameters of the i-th binary classifier

def one_vs_all_bin_theta(Z: NDArray) -> NDArray:
    theta_mat = np.zeros((p, g))
    for i in range(1, g + 1):
        # codebook in one vectorised step
        y_rec = np.where(Z[:, p] == i, 1.0, -1.0)
        Z_coded = np.column_stack([Z[:, :p], y_rec])
        theta_mat[:, i - 1] = minimize(
            emp_risk_bin_hinge,
            np.zeros(p),
            args=(Z_coded,)).x
    return theta_mat

# Prediction: pick class with max score.
def one_vs_all_bin_predict(theta_mat: NDArray,
                          x: NDArray) -> int:
    return int(np.argmax(x @ theta_mat)) + 1
```

Code (Python): multiclass (joint)

```
import numpy as np
from numpy.typing import NDArray
from scipy.optimize import minimize

# Vectorised empirical risk for the multiclass sum
# surrogate.
# Shapes: Theta is (p, g); scores (n, g); f_y (n,);
# margins (n, g).
# The k=y entry of 'margins' is always 1, so its max
# (0, .) = 1
# per row; subtract n_loc to recover the sum over k
# != y only.
def emp_risk_multiclass(Theta_flat: NDArray,
                        Z: NDArray) -> float:
    Theta = Theta_flat.reshape(g, p).T
    X, y = Z[:, :p], Z[:, p].astype(int)
    scores = X @ Theta
    n_loc = X.shape[0]
    f_y = scores[np.arange(n_loc), y - 1]
    margins = 1.0 + scores - f_y[:, None]
    return float(np.maximum(0.0, margins).sum() -
                  n_loc)
```

```
# Multiclass fitting: ONE joint optim over all p*g
# params.
# Output: Theta of shape (p, g). All columns are
# coupled
# by the joint optimisation.
def multiclass_theta(Z: NDArray) -> NDArray:
    Theta_flat = minimize(emp_risk_multiclass,
                          np.zeros(p * g),
                          args=(Z,)).x
    return Theta_flat.reshape(g, p).T

# Prediction: same argmax rule as one-vs-rest.
# Difference is only in how Theta is fit.
def multiclass_predict(Theta: NDArray,
                       x: NDArray) -> int:
    return int(np.argmax(x @ Theta)) + 1
```

Code (Python): setup, fit, compare on iris

```
import numpy as np
from sklearn.datasets import load_iris
# load iris and prepend an all-ones intercept column
iris = load_iris()
X = np.hstack([np.ones((iris.data.shape[0], 1)), iris.data])
y = iris.target + 1      # 1-based labels (parity w/ R)
Z = np.column_stack([X, y])
n, p = X.shape
g = int(len(np.unique(y)))
# train/test split (50 train, 100 test)
rng = np.random.default_rng(5)
train_ind = np.sort(rng.choice(n, size=50, replace=False))
test_ind = np.setdiff1d(np.arange(n), train_ind)
# fit both
theta_mat = one_vs_all_bin_theta(Z[train_ind])
Theta = multiclass_theta(Z[train_ind])
print(Theta)
# count agreements; print disagreements
count = 0
for j in test_ind:
    p_ovr = one_vs_all_bin_predict(theta_mat, Z[j, :p])
    p_mc = multiclass_predict(Theta, Z[j, :p])
    if p_ovr != p_mc:
        print(f"OvR: {p_ovr}, Multiclass: {p_mc}, "
              f"True: {int(Z[j, p])}")
    else:
        count += 1
print(count)
```

```
[[ 1.40942854  4.65434516 -6.0637737 ]
 [ 0.22520719  0.1572596  -0.38247367]
 [ 1.78485964  0.14583447 -1.93070101]
 [-0.96581534 -0.59320725  1.55903284]
 [-2.84883267 -1.59538011  4.44421608]]
```

```
OvR: 1, Multiclass: 2, True: 1
OvR: 1, Multiclass: 2, True: 2
98
```

98/100 test predictions agree — same data, same hypothesis class, but the two ERM's produce different fits.

Note

Numbers differ from R reference (95/100): different RNG and optim defaults. Qualitative finding unchanged.

Recap: link to the lecture material

Exercise takeaway	Lecture connection
(a) 0-1 loss $\leq$ multiclass hinge $L = \max_k (f_k(\mathbf{x}) - f_y(\mathbf{x}) + I_{y \neq k})$; case-split on whether $h(\mathbf{x}) = y$.	0-1 loss is discontinuous and NP-hard to optimise $\Rightarrow$ a continuous convex <i>surrogate</i> is needed; multiclass losses act on a score vector $f(\mathbf{x}) = (f_1, \dots, f_g)$.
(b) Hinge $\leq \sum_{k \neq y} \max\{0, 1 + f_k(\mathbf{x}) - f_y(\mathbf{x})\}$: replace $\max_k$ by a sum of clipped per-class margins.	Multiclass score-vector framework with the argmax predictor $h(\mathbf{x}) = \arg \max_k f_k(\mathbf{x})$.
(c) For $g = 2$, $f(\mathbf{x}) = f_1 - f_{-1}$: the sum surrogate reduces to the binary hinge $\max\{0, 1 - yf(\mathbf{x})\}$.	Binary hinge is the canonical convex 0-1 surrogate; zero exactly when the margin $yf(\mathbf{x}) \geq 1$.
(d) Surrogate = 0 $\iff$ all $g - 1$ margins $m_{y,k} = f_y(\mathbf{x}) - f_k(\mathbf{x}) \geq 1$ — confident-correct prediction generalised to multiclass.	Binary version (lecture): hinge is zero only past the margin — “no bonus for points beyond margin 1”.
(e) Joint multiclass ERM (one coupled optim over all $g - 1$ margins) vs. one-vs-rest (g independent binary ERMs) $\Rightarrow$ different fits on the same data.	One-vs-rest binary reduction: g classifiers fit independently, prediction by $\arg \max$; ERM as the empirical substitute for true-risk minimisation.